

SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME: VIMS (Vaccine integrity monitoring system)

GROUP NUMBER: Group 6

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Questions to identify measurements:

Schedule: Did the team finish building the IoT monitoring and AI alerting modules by the dates we planned?

Effort: How many total hours did we spend on the QR system compared to the time spent on hardware sensors?

Performance: Does the system successfully send 98% of the data and stay within 0.5°C of the real temperature

Reliability: Does the platform stay active for 99% of the time and send the notifications in less than 15 seconds?

Defects: How many major bugs did we find when connecting the sensors to the storage case?

Identified measurements:

Milestone Tracking: Planned vs. actual completion dates for the IoT, AI, and QR code milestones.

Weekly Effort: Total staff-hours spent per week on specific activities (ex: AI modeling, Hardware Integration).

Transmission Rate: Percentage of successful data packets transmitted compared to the total packets sent every 60 seconds.

Notification Latency: Number of seconds elapsed between a detected "Spoilage Risk" and the supervisor notification.

Defect Count: Total number of bugs identified, categorized by severity (Low, Medium, High, Critical).

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Measurement storage and collection:

What: Planned and real dates for milestones

- **When:** When each project part (AI, IoT, or QR) is finished.
- **Format:** Date format (DD/MM/YYYY).
- **How:** The team leader writes the dates into our shared project spreadsheet.

What: Weekly work hours for the team

- **When:** At the end of every week.
- **Format:** Number data (hours).
- **How:** Every team member records their own hours in a shared Google Sheet.

What: Data packet transmission rate

- **When:** Every 60 seconds while we are testing or running the system.
- **Format:** Percentage and integer counts.
- **How:** The server saves this data automatically in a system log file.

What: Seconds between alert and notification

- **When:** Each time we test a "breach probability" alert.
- **Format:** Real numbers (seconds).
- **How:** The system calculates the time automatically and the tester records it in the logs.

What: Number of bugs and their severity

- **When:** When someone finds a bug during the testing phase.
- **Format:** Integer counts.
- **How:** The person who finds the bug puts it into our GitHub Issues or a shared list.

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Measurement Type	Description	Example Measurements
Schedule	Tracks if the Cold Chain monitoring features were ready before the first physical container test.	Sensor integration: planned for week 1 completed in week 2
Effort	Measures the distribution of labor between software alerting and physical sensor calibration.	25 hours spent on AI Predictive Model training in Week 3
Performance	Measures the speed or accuracy of specific automated system features.	Alert Latency: 16 seconds from detection to supervisor notification.
Product Size	Measures the complexity of the VIMS system (ex: number of integrated sensors).	3 primary software modules built (IoT, AI, QR)

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Change	Monitors shifts in temperature or AI logic based on WHO vaccine safety standards.	2 updates made to the temperature range thresholds
Defects	Counts the number of errors or bugs found during the testing phases.	1 critical defect found in the AI module.
Quality	Measures how well the system meets its technical performance requirements.	Data Transmission Rate: 96% success rate achieved during testing.